

$$= \frac{1}{2} (102 + 112 - 122 + 132)$$

Step IV :- Reverse signs of all terms in $|x\rangle$ except $|0\rangle$
 $\text{OFT}_2 |x\rangle = \frac{1}{2} (102 - 112 + 122 - 132) = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$

Step V :- Apply inverse OFT_2^{-1} .

$$|x\rangle = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$|x\rangle = |2\rangle = |10\rangle$ Hence when we measure $|x\rangle$ we get solution is indexed at $x=2$

The factoring problem

Euclid's algorithm: The GCD is computed, recursively.

$$\text{gcd}(a, b) = \begin{cases} b & \text{if } a \bmod b = 0 \\ \text{gcd}(b, a \bmod b) & \text{otherwise} \end{cases}$$

Ex:- $\text{gcd}(18, 12) = \text{gcd}(12, 6) = 6$

$\text{gcd}(a, b) = 1$ if a and b are coprime.

$\text{gcd}(25, 16) = 1$

Period finding. Suppose a function $f(x) = f(x + r)$ then $f(x)$ is said to have a period r .

x	0	1	2	3	4	5	...
$f(x) = 11^x \bmod 15$	1	11	1	11	1	11	...

This function is of period 2.

Shor's Quantum factoring algo.

Step 1: choose a random integer $a \leq N$, such that

$\text{gcd}(a, N) = 1$

Step 2: Given Quantum algo. to compute period r .

Create 2 quantum states of required size & randomize them using H-gate. Find the period of $f(x) = a^x \bmod N$

Then using H-gate. Find the period of $f(x) = a^x \bmod N$

Step 3: If π is even & $a^{\pi/2} \bmod N \neq -1$, then compute $p = \gcd(N, a^{\pi/2} + 1)$ & $q = \gcd(N, a^{\pi/2} - 1)$ which are factors of N . otherwise go back to step 1 & select different 'a'.

Q1: Find the period π & hence find factors $N=15$ using Shor's algorithm.

Step 1: choose $a=11 < N=15$ with $\gcd(15, 11)=1$.

Step 2: Create 2 quantum states

state 1: $| \psi_1 \rangle$ with $k=3$ qubits

for representing the numbers from 0 to 7 ($\leq N/2$)

state 2: $| \psi_2 \rangle$ with $m=4$ qubits

for representing the numbers from 0 to 15 ($\leq N$).

Let $| \psi \rangle = | \psi_1 \rangle | \psi_2 \rangle$.

The size of $| \psi \rangle$ is 7 (3+4). Initialise all 7 qubits to zero 102.

$$| \psi \rangle = | 0000000 \rangle = \underbrace{| 000 \rangle}_{| \psi_1 \rangle} \underbrace{| 0000 \rangle}_{| \psi_2 \rangle}$$

Now apply H to each qubit of $| \psi_1 \rangle$.

$$| \psi \rangle = \left(\frac{1}{\sqrt{2}} (| 0 \rangle + | 1 \rangle) \right) \left(\frac{1}{\sqrt{2}} (| 0 \rangle + | 1 \rangle) \right) \left(\frac{1}{\sqrt{2}} (| 0 \rangle + | 1 \rangle) \right) | 0000 \rangle$$

$$= \frac{1}{\sqrt{8}} (| 000 \rangle + | 001 \rangle + | 010 \rangle + | 011 \rangle + | 100 \rangle + | 101 \rangle + | 110 \rangle + | 111 \rangle) | 0000 \rangle$$

$$= \left(\frac{1}{\sqrt{8}} \sum_{k=0}^7 | k \rangle \right) | 0000 \rangle$$

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$$f(x) = a^x \bmod N = 11^x \bmod 15$$

Evaluate $f(x) = 11^x \bmod 15$ for x in the state 1.

Simultaneously & store the result in state 2.

$$| \psi \rangle = \frac{1}{\sqrt{8}} (| 000 \rangle | 0000 \rangle + | 001 \rangle | 0010 \rangle + | 010 \rangle | 0100 \rangle + | 011 \rangle | 0110 \rangle + | 100 \rangle | 1000 \rangle + | 101 \rangle | 1010 \rangle + | 110 \rangle | 1100 \rangle + | 111 \rangle | 1110 \rangle)$$

$$\psi = \frac{1}{\sqrt{8}} \left(\begin{matrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{matrix} \right) \begin{matrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{matrix} + \begin{matrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{matrix} \begin{matrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{matrix}$$

- The state 1 contains the period τ of interest, but only for identical values in state 2.
- The period $\tau = 2$ is the distance between the components (0, 2, 4, 6 or 1, 3, 5, 7) in the state 1 for some value in state 2.

$C \propto 112$. $\tau = 2$ is even. $11 \bmod 15 \neq -1$.

The factors of $N=15$ are $p = \gcd(N, a^{\tau/2} + 1)$
 $= \gcd(15, 11 + 1) = \gcd(15, 12) = 3$.

~~$q = \gcd(15, 1)$~~

$q = \gcd(N, a^{\tau/2} - 1) = \gcd(15, 11 - 1)$
 $= \gcd(15, 10) = 5$.

The factors of $N=15 = p, q = 3, 5$.

2. Find the period for the f^n :

$f(x) = x^x \bmod 15$, using Shor's algorithm.

$a=7, N=15$.

Quantum Error Correcting Code (QECC)

To protect quantum states against the effects of noise, we would like to develop QECC.

difficulties in QECC:-

1. No cloning.
2. Errors are continuous.
3. Measurement destroys quantum information.

1. Three-qubit Bit Flip Code.

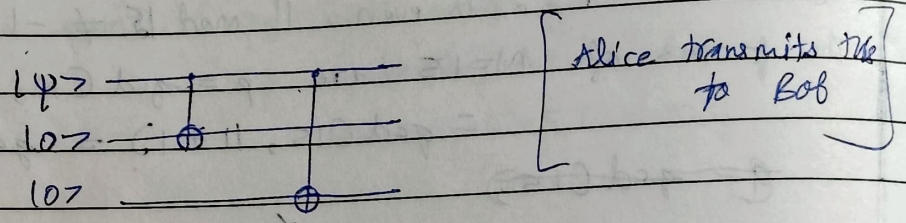
The single bit flip error is described by:- $10 \rightarrow 11$
 $11 \rightarrow 10$

The bit flip error is equivalent to action of X-gate.
Suppose Alice wants to send qubit $|\psi\rangle = a|0\rangle + b|1\rangle$ to Bob through a noisy quantum channel.

Encoding: Alice uses 3 qubits to encode one qubit. He will append 2 extra bits $|00\rangle$ to $|\psi\rangle$.

$$\begin{aligned} |\psi_1\rangle &= |\psi\rangle|00\rangle \\ &= (a|0\rangle + b|1\rangle)|00\rangle \\ &= a|000\rangle + b|100\rangle \end{aligned}$$

The circuit performing this encoding is :-

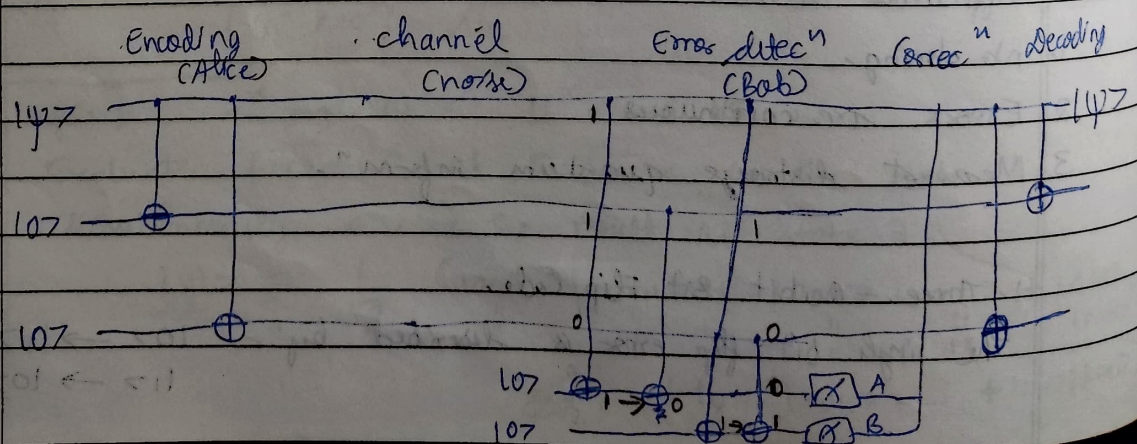


Error detection: Assuming that only a single qubit flip can occur during the transmission, Bob may receive one of the 4 possible states.

$$\begin{aligned} |\phi_0\rangle &= a|000\rangle + b|111\rangle \quad // \text{No error.} \\ |\phi_1\rangle &= a|001\rangle + b|110\rangle \quad // 1^{\text{st}} \text{ qubit flips} \\ |\phi_2\rangle &= a|010\rangle + b|101\rangle \quad // 2^{\text{nd}} \text{ " " " " } \\ |\phi_3\rangle &= a|100\rangle + b|011\rangle \quad // 3^{\text{rd}} \text{ " " " " } \end{aligned}$$

Now Bob has to extract from the received state which error occurred during the qubit transmission.

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Now how to extract from the received state which error occurred during the qubit transmission. Bob now introduces extra bits 100 as shown in figure, & applies 4 error operations.

Suppose $|\psi\rangle = a|001\rangle + b|110\rangle$ is received qubit by Bob. Then $C(a|001\rangle + b|110\rangle)100 = a|00100\rangle + b|110100\rangle$.

$S = a|00101\rangle + b|11001\rangle = (a|001\rangle + b|110\rangle)101$.

The output of extra bit is 101 (10), & it tells qubit error occurred in qubit position 1. Bob now corrects the error by applying X-gate to qubit position 1.

$$a|001\rangle + b|110\rangle \xrightarrow{X \text{ on qubit 1}} a|100\rangle + b|011\rangle$$

Now apply inverse encoding, to recover $|\psi\rangle$.

$$a|100\rangle + b|011\rangle \xrightarrow{\text{inverse encoding}}$$

$$a|100\rangle + b|011\rangle \xrightarrow{\text{inverse encoding}} a|000\rangle + b|111\rangle \xrightarrow{\text{inverse encoding}} a|00\rangle + b|11\rangle$$

3-qubit Phase flip code.

Phase flip QEC.

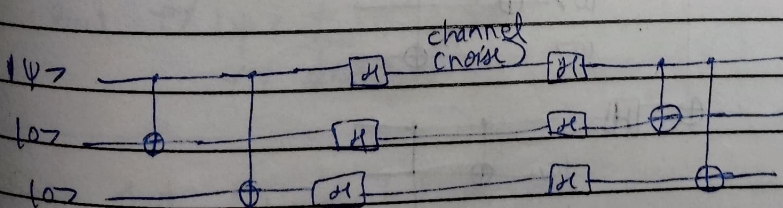
The phase flip error is equivalent to action of Z-gate.

$$a|0\rangle + b|1\rangle \xrightarrow{Z} a|0\rangle - b|1\rangle$$

Therefore phase flip error is $C(1 \rightarrow -1)$ is same as bit flip error $\rightarrow$

$$|+\rangle \rightarrow |-\rangle$$

$$|-\rangle \rightarrow |+\rangle$$



1. Append extra bits: $a|0\rangle + b|1\rangle = a|000\rangle + b|111\rangle$

2. Apply X-gate: $a|000\rangle + b|111\rangle \xrightarrow{X \text{ on all qubits}} a|111\rangle + b|000\rangle$

$$|111\rangle \rightarrow \frac{(107+117)}{\sqrt{2}} \frac{(107+117)}{\sqrt{2}} \frac{(107+117)}{\sqrt{2}} \cdot$$

them is, ~~as~~ $\rightarrow \frac{(10007+11117)}{\sqrt{2}}$

3. Apply bit flip algorithms in $(117, 1-7)$.
4. Apply H-gate again to recover original qubit.

⊕ Shor's nine qubit code.

It is a combinⁿ of the 3 qubit phase flip & bit flip codes.

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1. Encode the qubit using phase flip code.

$$a|07\rangle + b|17\rangle \rightarrow a|111\rangle + b|---\rangle$$

2. Encode each of these qubits using the three bit flip code, i.e., $|17\rangle$ is encoded as $\frac{(10007+11117)}{\sqrt{2}}$ &

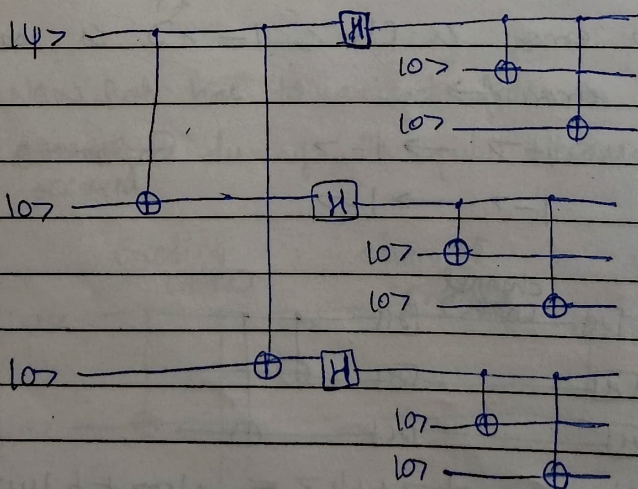
$$|-\rangle \text{ is encoded as } \frac{(10007-11117)}{\sqrt{2}}.$$

3. The result is a nine qubit code.

$$a \left[\frac{(10007+11117)(10007+11117)(10007+11117)}{2\sqrt{2}} \right] \rightarrow \text{results in 8 terms}$$

$$+ b \left[\frac{(10007-11117)(10007-11117)(10007-11117)}{2\sqrt{2}} \right] \rightarrow \text{results in 8 terms}$$

~~the~~



Imagine we encode $|\psi\rangle = a|0\rangle + b|1\rangle$ using Shor's 9 qubit code & then an X error occurs on the 8th qubit of the encoded state. Determine the state following the error.

$$a \begin{array}{cccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \uparrow & & & & & & & \\ 8^{\text{th}} & & & & & & & \end{array} |0000000\rangle + a \begin{array}{cccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \uparrow & & & & & & & \\ 8^{\text{th}} & & & & & & & \end{array} |0000001\rangle + \dots \text{etc.}$$

we get:-

$$a \left(\frac{(|010\rangle + |101\rangle)(|100\rangle + |111\rangle)(|100\rangle + |111\rangle)}{2\sqrt{2}} \right) +$$

$$b \left(\frac{(|010\rangle - |101\rangle)(|100\rangle - |111\rangle)(|100\rangle - |111\rangle)}{2\sqrt{2}} \right)$$

Stabilizer codes.

Stabilizer codes sometimes known as additive quantum codes. It is clearly illustrated by an example:-

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$\begin{array}{c} \begin{array}{cc} \times \times & \times \times \\ \uparrow \uparrow & \uparrow \uparrow \end{array} \\ \times \times |\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \\ = \frac{|11\rangle + |00\rangle}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\psi\rangle \end{array}$$

It is easy to write:-

$$\times \times |\psi\rangle = |\psi\rangle \quad \& \quad z z |\psi\rangle = |\psi\rangle$$

So we say that state $|\psi\rangle$ is stabilized by $\times \times$ & $z z$.

But $y y |\psi\rangle \neq |\psi\rangle$. So

$y y$ does not stabilize $|\psi\rangle$

$$\begin{array}{c} \begin{array}{cc} z z & z z \\ \uparrow \uparrow & \uparrow \uparrow \end{array} \\ z z |\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \end{array}$$

$$= \frac{|00\rangle + |-1-1\rangle}{\sqrt{2}}$$

$$= \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\psi\rangle$$

Show that the quantum state:-

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \text{ is stabilized by } \{I, Y\}$$

Solⁿ:- We have to show that :-

$$I|\psi\rangle = |\psi\rangle \quad \& \quad X|\psi\rangle = |\psi\rangle$$

Holds good
immediately

[First check if $|\psi\rangle$ is valid. Else
normalize]

$$I|\psi\rangle = \frac{1}{\sqrt{2}} (|10\rangle + i|01\rangle)$$

$$X|\psi\rangle = \frac{1}{\sqrt{2}} (X|10\rangle + X|i|01\rangle) = \frac{1}{\sqrt{2}} (|11\rangle + i|10\rangle)$$

$$= \frac{1}{\sqrt{2}} (i|11\rangle + |10\rangle) = \frac{1}{\sqrt{2}} (|10\rangle + i|11\rangle) = |\psi\rangle$$

Q2. Show that the quantum state :-

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \text{ is stabilised}$$

by $\{-XX, -ZZ\}$.

Solⁿ:- We have to show that $-XX|\psi\rangle = |\psi\rangle$

$$-ZZ|\psi\rangle = |\psi\rangle$$

$$-XX|\psi\rangle = - \left(\frac{\overset{XX}{|01\rangle} - \overset{XX}{|10\rangle}}{\sqrt{2}} \right) = \frac{|01\rangle - |10\rangle}{\sqrt{2}} = |\psi\rangle$$

$$-ZZ|\psi\rangle = - \left(\frac{\overset{ZZ}{|01\rangle} - \overset{ZZ}{|10\rangle}}{\sqrt{2}} \right) = - \left(\frac{-|01\rangle + |10\rangle}{\sqrt{2}} \right)$$

$$= \frac{|01\rangle - |10\rangle}{\sqrt{2}} = |\psi\rangle$$